Week 3

Spring 2027 · Section 017 · MW 7:30–8:50 PM · RRC Room 8210

Course Level Objectives:

- Explain the use of data collection and statistics as tools to reach reasonable conclusions. (CLO 1)
- Recognize, examine, and interpret the basic principles of describing and presenting data. (CLO 2)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Classify a study as observational or experimental. (CLO 1) **2.** Identify a plausible confounding variable in an observational study. (CLO 1) **3.** Explain how random assignment, a control group, and blinding support cause-and-effect conclusions. (CLO 1) **4.** Distinguish random sampling from random assignment. (CLO 1)	e-Text: 1.3 Experiments and Observational Studies Interactive Lecture: 1.3 Experiments and Observational Studies Slides: 1.3 Experiments and Observational Studies (includes TED-Ed placebo video)	Participation: Experiments and Observational Studies – classify studies; confounding; random sampling vs. random assignment Your Turn: Study Design and Scope of Inference (coffee and depression; vitamin D experiment)	Online Homework: 1.3 Experiments and Observational Studies (due Sun Feb 7, 11:59 PM) Take-Home Quiz 1 (due Mon Feb 8, 11:59 PM) – covers Intro, 1.1–1.3 (all objectives) Test 1 (Fri Feb 26 – Sun Feb 28, proctored) – covers Intro, 1.1–2.5 (all objectives) Test 1 Study Guide (all objectives) Practice Test 1 (optional; due Thu Feb 25) (all objectives)	WileyPLUS StatKey Blackboard Respondus
5. Construct frequency and relative frequency tables, compute a sample proportion $\hat{p}$, and distinguish the parameter p from the statistic $\hat{p}$. (CLO 2) **6.** Compute and interpret overall and conditional proportions, and the difference in proportions, from a two-way table. (CLO 2) **7.** Interpret bar charts, pie charts, side-by-side bar charts, and segmented bar charts. (CLO 2)	e-Text: 2.1 Categorical Variables Interactive Lecture: 2.1 Categorical Variables Slides: 2.1 Categorical Variables	Participation: Frequency and Relative Frequency Participation: Finding Proportions from Two-Way Tables Your Turn: Two-Way Table Proportions (pet ownership by housing)	Online Homework: 2.1 Categorical Variables (due Sun Feb 7, 11:59 PM) Take-Home Quiz 2 (due Mon Feb 15, 11:59 PM) – covers 2.1–2.3 (all objectives) Test 1 (Fri Feb 26 – Sun Feb 28, proctored) – covers Intro, 1.1–2.5 (all objectives) Test 1 Study Guide (all objectives) Practice Test 1 (optional; due Thu Feb 25) (all objectives)	WileyPLUS StatKey Blackboard Respondus